

Worksheet 4B: What Derivatives and Integrals Tell Us

AIML Foundations Mathematics

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What This Worksheet Is About We've learned how to compute derivatives and integrals. Now: what do they mean? - **Derivatives** → rate of change, slope, velocity, increasing/decreasing - **Integrals** → accumulation, area, total distance, total quantity These interpretations are why calculus is so powerful!

Part A: Derivatives — Rate of Change and Slope (16 problems)

Derivative	Meaning
$f'(x) > 0$	Function increasing ↗
$f'(x) < 0$	Function decreasing ↘
$f'(x) = 0$	Horizontal tangent (potential max/min)
$f''(x) > 0$	Concave up ∪
$f''(x) < 0$	Concave down ∩

- For $f(x) = x^2$, $f'(x) = 2x$. - (a) Where is f increasing? Decreasing? - (b) Where is the tangent horizontal? - (c) What is the slope at $x = 3$?
- For $g(x) = x^3 - 3x$: - (a) Find $g'(x)$. - (b) Find critical points (where $g'(x) = 0$). - (c) Classify as local max or min.
- For $h(t) = t^4 - 8t^2$: - (a) Find all critical points. - (b) Use the first or second derivative test to classify each.
- Find the tangent line to $f(x) = x^2$ at $(2, 4)$.
- Find the tangent line to $g(x) = x^3 - x$ at $x = 1$.
- At what point(s) on $y = x^3 - 12x$ is the tangent horizontal? **Physics Applications:**
- Height: $h(t) = 20t - 5t^2$ meters. - (a) Find velocity $v(t) = h'(t)$. - (b) When is velocity zero? What's happening then? - (c) Find acceleration $a(t) = h''(t)$. Interpret it.
- Position: $s(t) = t^3 - 6t^2 + 9t$ meters. - (a) Find velocity and acceleration. - (b) When is the object at rest? - (c) When is acceleration zero? **Second Derivative and Concavity:**
- For $f(x) = x^3$:
 - Where is $f''(x) > 0$? (Concave up)

- Where is $f''(x) < 0$? (Concave down)
 - Where is the inflection point?
10. For $g(x) = x^4 - 6x^2$:
- Find inflection points.
 - Describe concavity on each interval.
- Optimization:**
11. Profit: $P(x) = -x^2 + 80x - 1200$.
- Find the quantity maximizing profit.
 - Find maximum profit.
12. Area with fixed perimeter: A rectangle has perimeter 100m.
- Express area as $A(w) = w(50 - w) = 50w - w^2$.
 - Find dimensions maximizing area.
13. A ball's height is $h(t) = -5t^2 + 30t + 10$.
- When does it reach maximum height?
 - What is the maximum height?
14. Revenue: $R(p) = 500p - 2p^2$.
- Find price maximizing revenue.
 - Find maximum revenue.
15. Cost: $C(q) = q^3 - 12q^2 + 60q + 100$.
- Find marginal cost $C'(q)$.
 - At what quantity is marginal cost minimized?
16. Summary: The derivative being zero tells us _____, and the second derivative's sign tells us _____.

Part B: Integrals — Accumulation and Area (16 problems)

Integral	Meaning
$\int_a^b f(x) dx$	Signed area under curve
$\int v(t) dt$	Displacement
$\int r(t) dt$	Total quantity accumulated

Area Under Curves:

17. Find the area under $f(x) = 3$ from $x = 1$ to $x = 5$.

- Geometrically: rectangle with base _____ and height _____
 - By integral: $\int_1^5 3 \, dx =$
18. Find the area under $f(x) = x$ from $x = 0$ to $x = 4$.
- Geometrically: triangle
 - By integral: $\int_0^4 x \, dx =$
19. $\int_0^2 x^2 \, dx =$
20. $\int_1^3 (2x + 1) \, dx =$ **Signed Area:**
21. $\int_{-2}^2 x \, dx =$ _____ Why? (Think about symmetry)
22. $\int_{-1}^1 x^3 \, dx =$ _____ This is an odd function on a symmetric interval.
23. $\int_{-2}^2 x^2 \, dx =$ _____ This is an even function: $\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$
24. For $f(x) = x - 1$ from $x = 0$ to $x = 2$:
- Sketch the function.
 - Where is it positive? Negative?
 - Calculate $\int_0^2 (x - 1) \, dx$. Interpret the result.

Physical Applications:

25. Velocity: $v(t) = 3t^2$ m/s. Find displacement from $t = 0$ to $t = 2$.
26. Velocity: $v(t) = t - 2$ m/s from $t = 0$ to $t = 4$.
- When is the object moving forward? Backward?
 - Find net displacement: $\int_0^4 (t - 2) \, dt$
 - Find total distance traveled.
27. Flow rate: $r(t) = 100 - 4t$ liters/min.
- Find total flow from $t = 0$ to $t = 10$.
28. Population growth rate: $P'(t) = 200 + 10t$ people/year.
- Find total growth over 5 years.

Accumulation Functions:

29. Let $A(x) = \int_0^x t^2 \, dt$.
- Compute $A(x)$ explicitly.
 - Find $A(1)$, $A(2)$, $A(3)$.
 - Find $A'(x)$. What do you notice?
30. If $F(x) = \int_1^x (t^3 + 1) \, dt$, what is $F'(x)$? (Use the Fundamental Theorem — no need to compute the integral!)

31. If $G(x) = \int_0^x \sqrt{t} dt$, what is $G'(x)$?
32. Explain: "Integrating then differentiating gets you back to where you started."

Part C: The Fundamental Theorem — Tying It Together (12 problems)

The Fundamental Theorem of Calculus:

1. $\frac{d}{dx} [\int_a^x f(t) dt] = f(x)$
 2. $\int_a^b f(x) dx = F(b) - F(a)$ where $F' = f$
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Using Part 1:

33. $\frac{d}{dx} [\int_0^x t^4 dt] =$
34. $\frac{d}{dx} [\int_2^x (3s^2 - s) ds] =$
35. $\frac{d}{dx} [\int_1^x \frac{1}{t^2} dt] =$ **Using Part 2:**
36. $\int_0^3 2x dx =$ (use antiderivative x^2)
37. $\int_1^4 (x^2 - 1) dx =$
38. $\int_0^8 x^{1/3} dx =$
39. $\int_1^e \frac{1}{x} dx =$ (antiderivative is $\ln x$; we'll derive this in 4D!) **Round-Trip Verification:**
40. Start with $f(x) = x^3$.
 - Integrate: $\int f(x) dx =$
 - Differentiate your answer.
 - Did you get $f(x)$ back?
41. Start with $g(x) = 6x^2 - 4x$.
 - Differentiate: $g'(x) =$
 - Integrate your answer.
 - Did you get $g(x)$ back (plus C)?
42. Explain the asymmetry:
 - Differentiating then integrating gives $f(x) + C$
 - Integrating then differentiating gives exactly $f(x)$

Why the difference? **Connecting Position, Velocity, Acceleration:**

43. Complete the chain: Position $s(t)$ $\xrightarrow{\text{derivative}}$ _____ $\xrightarrow{\text{derivative}}$ _____
 _____ $\xrightarrow{\text{integral}}$ Velocity
 _____ $\xrightarrow{\text{integral}}$ _____

44. An object has acceleration $a(t) = 6t$, with $v(0) = 2$ and $s(0) = 1$.
- Find $v(t)$ by integrating $a(t)$ and using the initial condition.
 - Find $s(t)$ by integrating $v(t)$ and using the initial condition.

Part D: Area Between Curves (8 problems)

Area between $f(x)$ and $g(x)$ from a to b :

$$\text{Area} = \int_a^b |f(x) - g(x)| dx$$

If $f(x) \geq g(x)$ on $[a, b]$: $\text{Area} = \int_a^b (f(x) - g(x)) dx$

45. Find area between $y = x^2$ and $y = x$ from $x = 0$ to $x = 1$.
- Which is on top?
 - Set up and compute the integral.
46. Find area between $y = x^2$ and $y = 4$.
- Find intersection points.
 - Set up and compute.
47. Find area enclosed by $y = x^2$ and $y = 2x$.
- Find intersections.
 - Integrate from intersection to intersection.
48. Find area between $y = x^3$ and $y = x$ from $x = -1$ to $x = 1$.
- Careful: which is on top changes at $x = 0$!

Applications:

49. Consumer Surplus: Demand is $p = 50 - 2q$, equilibrium price is $p^* = 30$.
- Consumer surplus $= \int_0^{q^*} (50 - 2q - 30) dq$
 - Find q^* and calculate surplus.
50. The area between a velocity curve and the t-axis represents _____.
51. The area between two position curves represents the _____ between them.
52. Gini Coefficient: Perfect equality is $y = x$. A Lorenz curve is $y = x^2$.
- $\text{Gini} = \frac{\text{Area between curves}}{\text{Area under } y=x}$
 - Calculate it.

Part E: Connecting to ML and Optimization (8 problems)

53. Loss function: $L(w) = w^2 - 6w + 10$.
- Find $L'(w)$.
 - Find the minimum.
 - Verify with second derivative.
54. Gradient descent update: $w_{new} = w_{old} - \alpha \cdot L'(w_{old})$ For $L(w) = w^2 - 4w + 5$ with $\alpha = 0.1$, starting at $w_0 = 0$: - Calculate w_1, w_2, w_3 . - What value is w approaching?
55. Total loss over dataset $[0, 4]$ with loss function $\ell(x) = (x - 2)^2$: Total = $\int_0^4 (x - 2)^2 dx$. Calculate it.
56. Mean loss = $\frac{1}{4} \int_0^4 (x - 2)^2 dx$. Calculate it.
57. Regularized loss: $L(w) = (w - 5)^2 + 0.5w^2$.
- Expand and simplify.
 - Find optimal w .
 - Compare to unregularized optimum $w = 5$.
58. Explain: Why is finding where $L'(w) = 0$ central to training ML models?
59. Explain: Why might we care about $\int L(w)dw$ over a region of weight space?
60. The Fundamental Theorem connects local information (derivative at a point) to global information (integral over an interval). How does this relate to:
- Local vs. global minima in optimization?
 - Gradient (local) vs. total loss (global)?